Backtest statistics: calculation guide

Scope and return convention

Statistics are calculated independently for each strategy and reporting period. Daily strategy return r = position held during the day x futures daily return. Returns include flat days; transaction costs are not included. The 2026 sample ends on 2 November 2026. Annualization uses 252 trading days per year.

Strategy, dates, year and observations

strategy identifies the model and whether CDF position sizing is volatility-scaled. start_date and end_date are the first and last observations in the summary period. year identifies the calendar-year period in the annual report. observations counts all daily rows in the period, including flat and zero-return days.

First active date and average absolute position

first_active_date is the first day with a nonzero position held during the period. average_abs_position is the mean absolute position over all period observations, including flat days. It measures average exposure, not return or volatility.

Total return

total_return_pct = 100 x (product of (1 + daily return) - 1).
This is the compounded return over the available reporting period, not annualized.

Annualized return and volatility

annualized_return_pct = 100 x mean daily return x 252. This is an arithmetic annualization, not a compounded annual growth rate (CAGR).
annualized_volatility_pct = 100 x sample standard deviation of daily returns x square root of 252. Sample standard deviation uses N - 1 in its denominator.

Sharpe ratio

sharpe_ratio = mean daily return / daily return standard deviation x sqrt(252).
The calculation assumes a zero risk-free rate and includes zero-return days.
It is undefined if daily return volatility is zero or not finite.

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Maximum drawdown

Wealth is the cumulative product of (1 + daily return). For each observation,
drawdown = wealth / running maximum wealth - 1. max_drawdown_pct is
100 x the most negative observed drawdown, expressed as a negative percentage.
The running peak begins with the first observed wealth, not a separate pre-period
wealth observation. Each annual report period restarts the wealth calculation.

Calmar ratio

calmar_ratio = arithmetic annualized return / absolute maximum drawdown.
Both quantities use the same reporting period; the annualized return is not CAGR.
If maximum drawdown is zero, the ratio is undefined and the CSV cell is blank.

Longest drawdown duration

max_drawdown_duration_calendar_days is the longest time in calendar days from
a running wealth peak to the first observed recovery to that peak or higher.
If not recovered by period end, the final observation ends the measurement.
The longest episode need not contain the deepest drawdown. It is not trading days.

Win ratio

win_ratio_pct = 100 x positive-return days / nonzero-return days.
Zero-return days, including inactive days, are excluded from the denominator.
If all days have zero returns, the ratio is undefined and the CSV cell is blank.

Risk-reward ratio

risk_reward_ratio = average positive daily return / absolute average negative
daily return. Zero-return days are excluded from both averages. The statistic
is based on days, not trades; it is undefined if either positive or negative days
are absent, and the CSV cell is blank.

Example

Six daily returns: +0.2%, +0.1%, +0.3%, -0.1%, -0.3%, 0%.
Win ratio = 3 / 5 = 60%. Average gain = 0.2%; average loss magnitude = 0.2%.
Risk-reward ratio = 0.2 / 0.2 = 1.0.

Trading volume

trading_volume = number of days the position changes in the reporting period.
A size change or a reversal counts once, regardless of its magnitude.
At the start of a backtest, the first position is compared with zero (flat).
Annual counts compare the first day with the prior trading day's position;
a position carried across a year boundary is not counted unless it changes.

Backtest statistics: CAGR supplement

Compound annual growth rate (CAGR)

CAGR (%) = 100 x (ending wealth $^{(365.25 / \text{elapsed calendar days})}$ - 1).

Ending wealth is the product of (1 + daily strategy return) over the reporting period, starting from wealth of 1. Elapsed calendar days is the difference between the last and first observation dates in the period.

CAGR is undefined if elapsed calendar days are zero or negative, or if ending wealth is zero or negative.

The recent six-model backtests use net daily strategy returns after trading costs. The historical backtests described in the preceding pages exclude trading costs; applying this formula to those saved returns therefore yields a gross CAGR.

CAGR is a compounded rate and differs from the arithmetic annualized return, which is the mean daily strategy return multiplied by 252 and expressed as a percentage. A CAGR calculated from a partial-year period is annualized, not the return actually earned over a full year.